

C S C E 6 7 6 — D A T A M I N I N G · F I N A L P R O J E C T

Citations, Communities, and Categories

What does citation structure tell us about a paper's topic?

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Dataset: OGBN-Arxiv — Open Graph Benchmark

169,343
papers

1,166,243
citations

*"A citation is a minimal signal — one paper pointing to another.
Does that signal, stripped of any text, already know what a paper is about?"*

Q1

Does structure know the topic?

If I hand you a citation graph with every title, abstract, and author scrubbed off, can you still tell me what field each paper belongs to? If "mostly yes," a large fraction of content classification can run on graphs alone — critical for cold-start cases where text is unreliable or missing.

Q2

Given text, does structure add anything?

Text embeddings are cheap and ubiquitous. If they already do 95% of the job, complex graph models are overkill. If they do 70% and a GNN pushes it to 90%, the business case flips. This sets the bar for expensive graph-aware infrastructure.

→ *These questions decide whether graph-based infrastructure is worth building — and feed directly into fairness debates about citation-based ranking.*

Research Questions

§ 2

MAIN QUESTION

For predicting the arXiv subject of a CS paper, how much signal lives in the citation structure alone, and how much lift does a GNN add by combining structure with text?

+ RQ - C · MECHANISM Does class-level homophily explain class-level GNN lift? — closes the loop between A and B (see §11).

RQ - A · COURSE METHOD

Unsupervised structural recovery

Can Louvain community detection, applied to the undirected citation graph with no access to text or labels, recover the 40 arXiv subject categories?

METRIC Normalized Mutual Information (NMI) between communities and labels; majority-label classifier accuracy.

RQ - B · BEYOND COURSE

Structure + features via a GNN

Does a GraphSAGE GNN — which aggregates neighbor features over the citation structure — outperform a text-only MLP baseline on held-out future papers?

METRIC Macro-F1 on the official OGB time-split; dissected by class and by test year.

Dataset — OGBN-Arxiv

§ 3

Open Graph Benchmark • Node Property Prediction

A directed citation graph of arXiv CS papers.

Nodes (papers)	169,343
Directed edges (citations)	1,166,243
Node features	128-d text embedding
Labels	40 arXiv CS subject areas
Publication years	1971 – 2020
License	ODC-BY

TIME-AWARE SPLIT

TRAIN • papers $\leq$ 2017 (90,941)

VALID • papers = 2018 (29,799)

TEST • papers $\geq$ 2019 (48,603)

The test papers did not exist when training data was collected — this rules out information leakage and simulates the real deployment setting for a paper-categorization system.

CITATION HOMOPHILY 65.5% of edges connect same-class papers — 26.2× random (2.5%). This is the premise of everything that follows.

Four EDA Findings That Shape the Pipeline

§ 4

Every finding below drives a specific modeling choice. Anything that did not was pruned.

01

HEAVY-TAILED DEGREE

6.9 / 1 / 13,155

Mean / median / max in-degree. The "average" paper is fictional; 36.6% of papers have zero in-degree inside this graph.

Consequence → use sparse-matrix ops; hubs dominate centrality and GNN gradient flow.

02

SEVERE CLASS IMBALANCE

942× ratio

Largest class = 27,321 papers, smallest = 29. Top-5 classes alone hold 53.5% of all papers.

Consequence → Macro-F1 (not accuracy) is the headline metric.

03

TEMPORAL SKEW

2017 VS 2015

Global median year is 2017, but the top-100 papers by in-degree have median year 2015. Test set lives in the steep-growth era.

Consequence → time-aware split is non-negotiable; centrality is age-biased.

04

CITATION HOMOPHILY

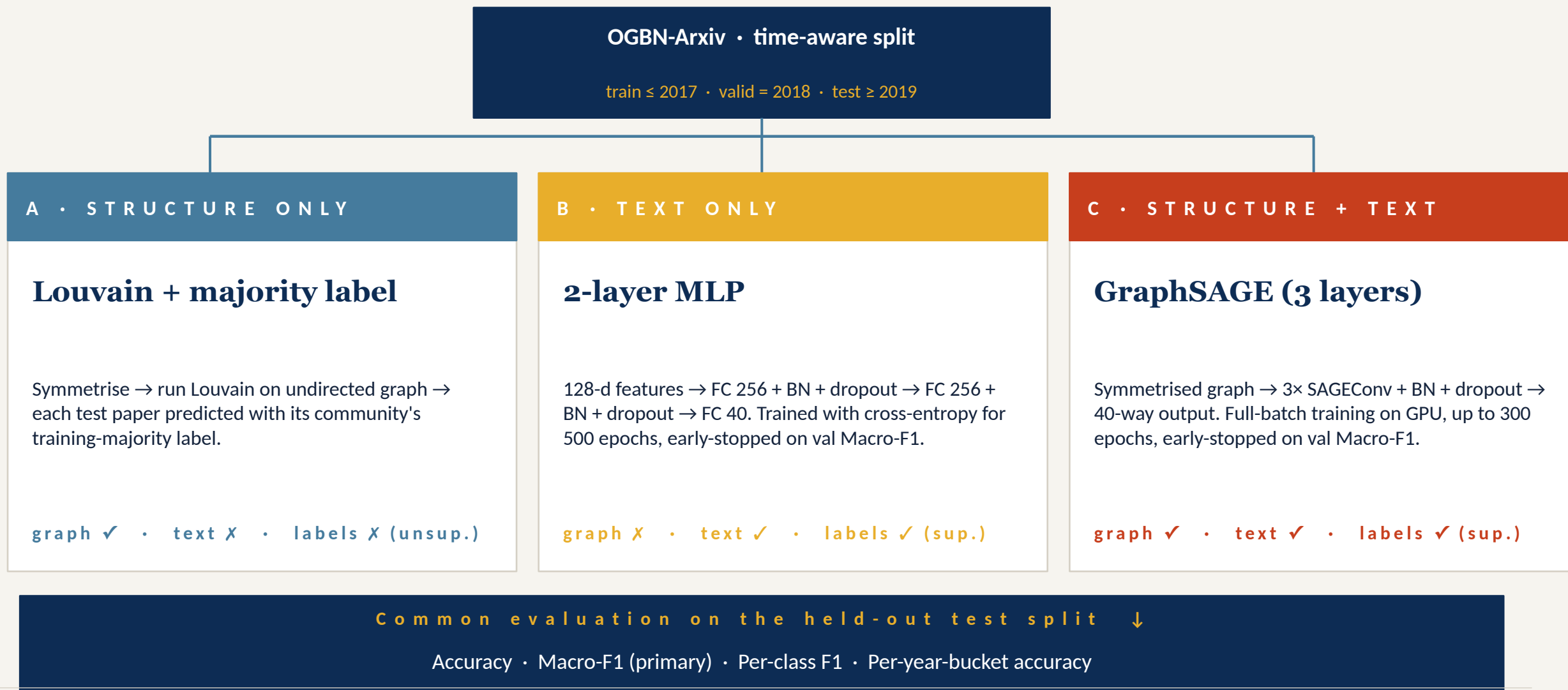
65.5%

763,986 of 1,166,243 citations connect papers with the same label — 26.2× above the 2.5% random baseline.

Consequence → structure carries real topical signal; GNN aggregation will work.

Methodology — Three Models, One Benchmark

§ 5



Approach A — Structure only (Louvain)

§ 6 · R Q - A

METHOD

The raw signal in citations, before any label sees any feature.

1. Symmetrise the directed citation graph (Louvain needs undirected).
2. Run Louvain community detection (modularity maximisation).
3. Evaluate the partition against labels with NMI and modularity.
4. Promote the partition to a classifier: majority training label per community.

INTERPRETATION

Modularity = 0.7020 → real large-scale community structure. NMI = 0.3853 → a meaningful fraction of that structure lines up with the 40-way arXiv taxonomy. You could build a usable paper classifier with nothing but the citation graph.

0.7020

MODULARITY

Far above the 0.3 'real community' threshold.

0.3853

NMI vs. GROUND TRUTH

Random = 0, perfect = 1. Clustering is unsupervised.

47.21%

MAJORITY-LABEL ACCURACY

Test accuracy with zero access to text. Random baseline = 2.5% (1/40).

Approach B — Text only (MLP)

§ 7

METHOD

The honest comparison point for the GNN.

- 2-hidden-layer MLP, hidden 256, BN + dropout 0.5.
- Adam, lr = 0.01, weight decay = 5e-4.
- Cross-entropy loss, up to 500 epochs.
- Early stopping on validation Macro-F1.
- Input = 128-d skip-gram text embedding. No graph.

NUANCE

The MLP cleanly beats Louvain on both metrics — 56.12% vs. 47.21% accuracy and 0.3301 vs. 0.1875 Macro-F1. Skip-gram text features are doing real work even before the graph; structure-only majority voting is no longer the better classifier.

56.12%

TEST ACCURACY

Text features only; no citations seen.

0.3301

TEST MACRO-F1

Unweighted mean per-class F1 — fairer under heavy class imbalance than plain accuracy.

Approach C — Structure + Text (GraphSAGE)

§ 8 · R Q - B

METHOD

Same text features as the MLP — plus the graph.

- 3 × SAGEConv layers, hidden 256, BN + dropout 0.5.
- Symmetrised graph: messages flow cited ↔ citing.
- Full-batch GPU training (fits on Colab T4).
- Adam, lr = 0.01, up to 300 epochs, early stop on val Macro-F1.
- Inductive design — natural fit with time-aware split.

WHY IT WINS

*65.5% homophily means neighbours usually share the label; aggregation is label smoothing.
Depth-3 message passing lets thin-text papers borrow context from richer neighbours.*

70.93%

TEST ACCURACY

Same 128-d features as the MLP — the only change is adding the graph.

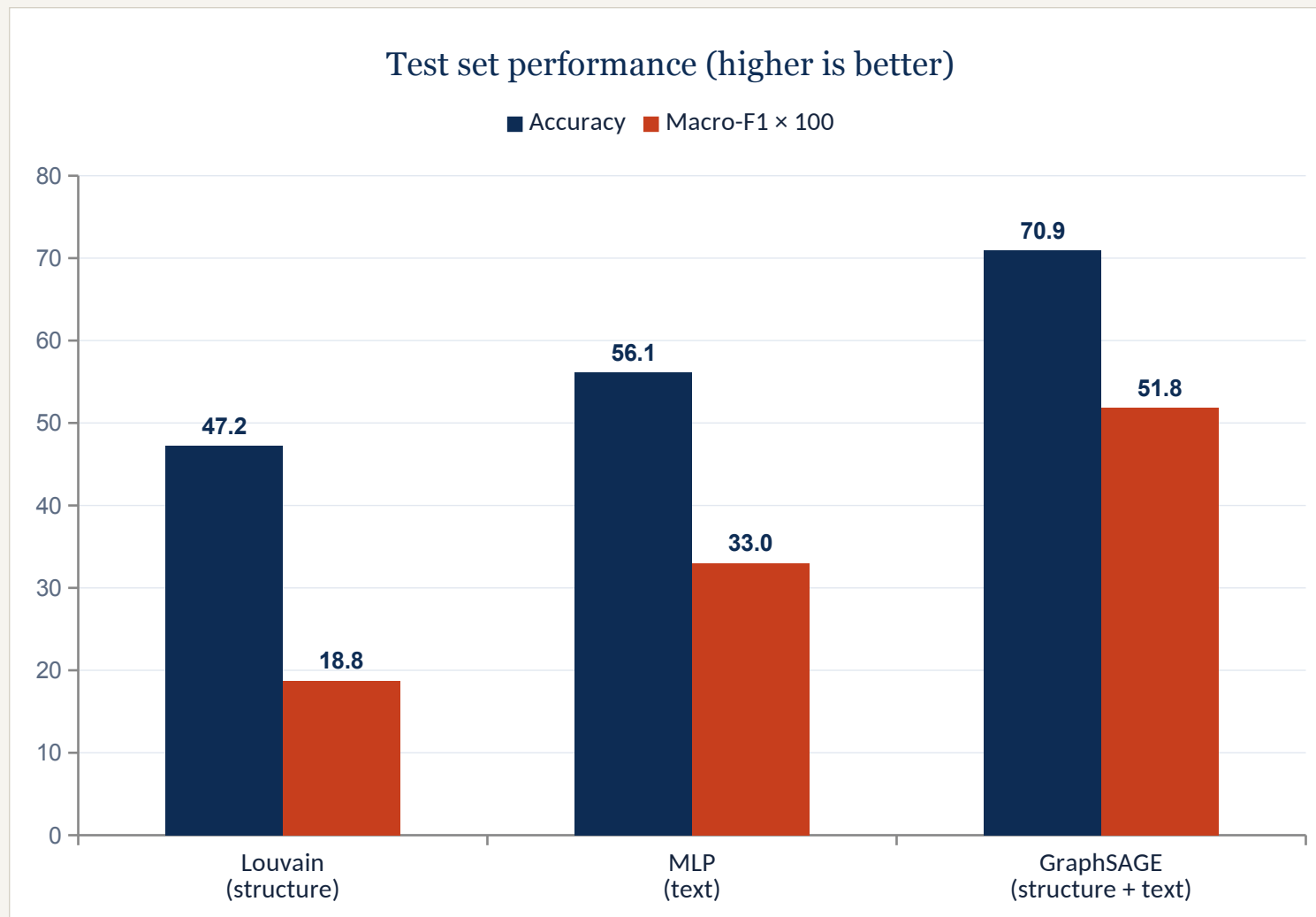
0.5183

TEST MACRO-F1

The jump in Macro-F1 confirms the lift is not just on majority classes — minority classes benefit too.

Results — Structure vs. Text vs. Both

§ 9



THE HEADLINE

Graph lift over text-only

+14.8

percentage points of accuracy

+18.8

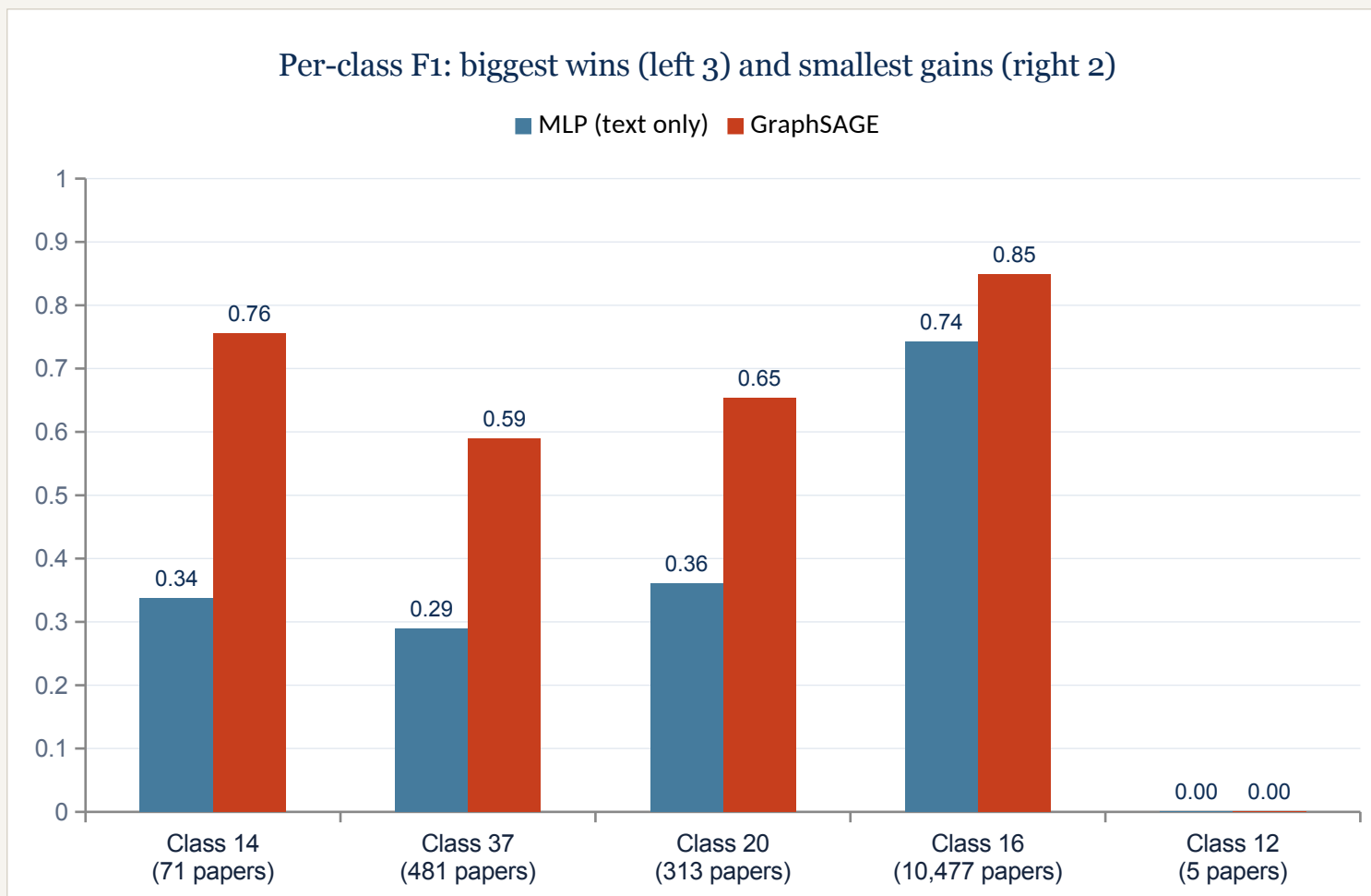
points of Macro-F1

The GNN sees exactly the same 128-d features as the MLP — the lift is pure graph signal.

Where Does the Graph Help Most?

§ 10

Per-class F1 lift (GraphSAGE – MLP)



WINS ↑

Graph context rescues classes where text is sparse or ambiguous:

- **Class 14:** F1 0.337 → 0.756 (71 papers, $\Delta +0.419$)
- **Class 37:** F1 0.289 → 0.589 (481 papers, $\Delta +0.300$)
- **Class 20:** F1 0.361 → 0.653 (313 papers, $\Delta +0.292$)

SMALLEST GAINS ◦

No class regressed in this run — the two cases below are flat or already-strong:

- **Class 16:** F1 0.742 → 0.849 (10,477 papers, $\Delta +0.107$)
- **Class 12:** F1 0.000 → 0.000 (only 5 test papers — statistically silent)

When the MLP is already strong (large classes) or there is nothing to learn ($n=5$), there is little room for the graph to add.

Why Does the Graph Help Where It Does?

RESEARCH QUESTION C

Is GraphSAGE's per-class lift over the MLP mechanically driven by per-class graph homophily?

METHOD

For each class $c \in \{0, \dots, 39\}$ we compute:

$H(c)$ = fraction of citation-edge endpoints in class c that point to another class- c paper.

$\Delta F1(c)$ = $F1_{GNN}(c) - F1_{MLP}(c)$ (already computed in the per-class breakdown).

Test: Spearman rank correlation across all 40 classes.

MECHANISM Message passing helps where neighbors carry information about a node — i.e., where citations stay within-class. If the per-class lift correlates with per-class homophily, RQ-A's structural premise mechanically explains RQ-B's modeling lift.

+0.424

SPEARMAN ρ ($p = 0.0065$)

Per-class homophily $H(c)$ vs. per-class GNN lift $\Delta F1(c)$ across 40 classes.

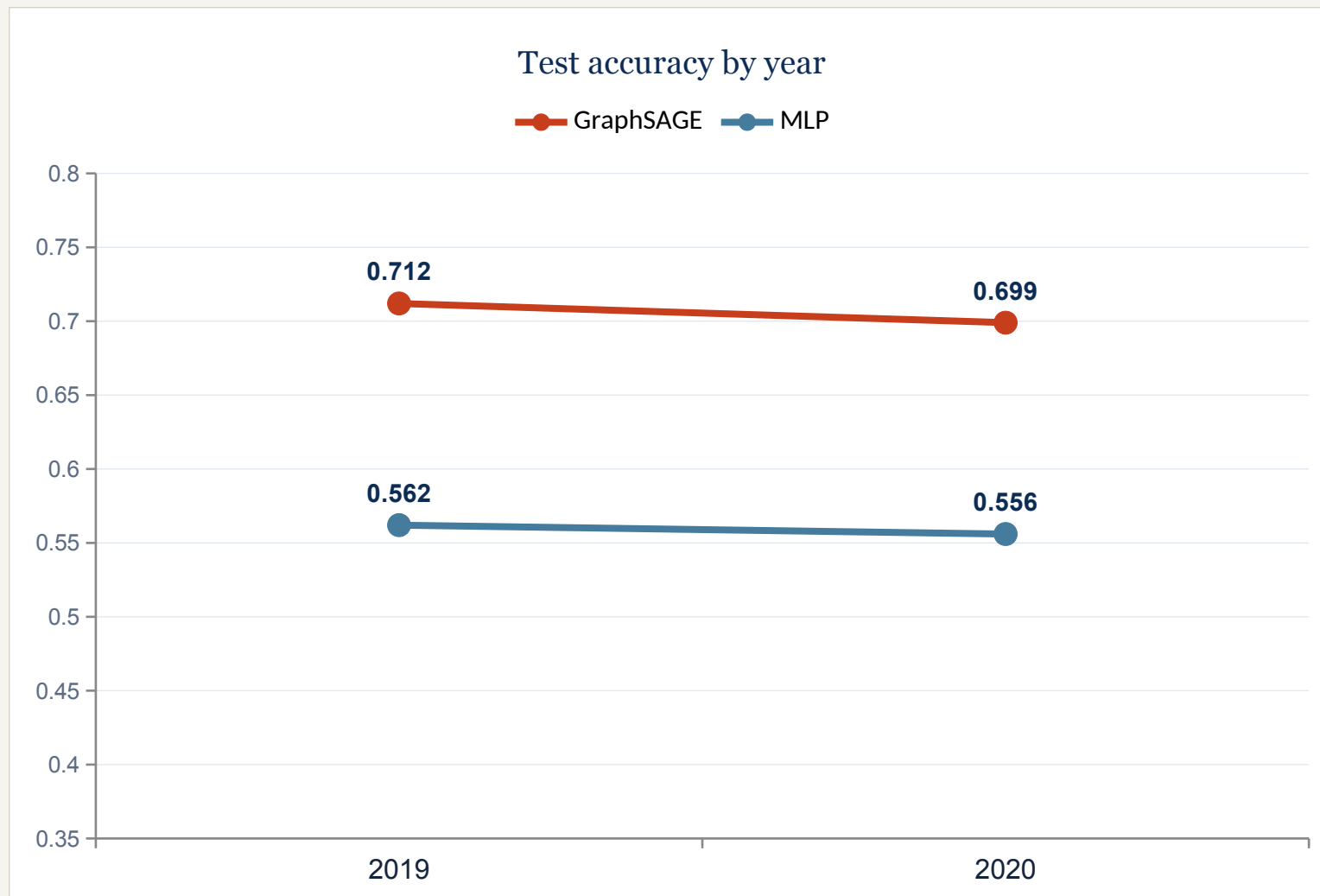
+0.437

PEARSON r ($p = 0.0048$)

Linear sanity check — both correlations are positive and significant at the 1% level.

Temporal Drift — Is the Lift Stable?

§ 1 2



OBSERVED

Both models drop modestly from 2019 to 2020.

MLP: 0.562 → 0.556 accuracy
0.330 → 0.323 Macro-F1

GraphSAGE: 0.712 → 0.699 accuracy
0.520 → 0.501 Macro-F1

THE TAKEAWAY

The lift is not a single-year artefact. GraphSAGE sits well above the MLP in both years. Mild drift is expected — test papers have fewer training-era neighbours to aggregate from.

Aside — Centrality Is Not Neutral

§ 1 3

PageRank and in-degree are highly correlated (Spearman $\rho = 0.9254$) — but both are heavily age-biased.

2017

GRAPH-WIDE MEDIAN YEAR

All 169,343 papers.

2016

TOP-500 BY IN-DEGREE

Median year — 1 year older than the graph-wide median.

2011

TOP-500 BY PAGERANK

Median year — 6 years older. PageRank is much more skewed toward early papers.

WHY THIS MATTERS FOR ANY SYSTEM BUILT ON THESE METRICS

- A recommender surfacing "highly-cited foundational work" structurally favours older authors and ideas.
- Paper-scoring systems used in hiring / grants / promotions risk encoding this bias.
- Neither in-degree nor PageRank is a **quality** measure. Both are **exposure-over-time** measures.

Ethics, Limitations, Threats to Validity

§ 1 4

1

Label is a simplification

arXiv assigns one primary category per paper; many papers are genuinely interdisciplinary. Our metrics reward guessing only the primary tag.

2

Centrality bias

Both in-degree and PageRank favour older papers. Not used as GNN features, but any downstream ranking/scoring extension must contend with this.

3

Dataset scope

OGBN-Arxiv covers arXiv CS only. Homophily and graph-lift conclusions are specific to this network.

4

The graph is the view we were given

36.6% of nodes have in-degree 0 — an artefact of the dataset boundary, not a statement about those papers. Their centrality and neighbour-quality are downweighted.

5

Class imbalance is not fully solved

Smallest classes are statistically shaky regardless of architecture; we cannot cleanly distinguish model weakness from tiny sample sizes.

6

We did not tune aggressively

A serious hyperparameter sweep would likely raise absolute scores on both MLP and GraphSAGE; treat our deltas as representative, not maximal.

Conclusion

Three tightly-coupled questions, three concrete answers.

R Q - A • S T R U C T U R E O N L Y

Modularity = 0.7020, NMI = 0.3853 → Louvain recovers a real, label-aligned community structure. A bare majority-label classifier reaches **47.21% accuracy** without ever touching a text feature. *Citation structure knows a lot about topic.*

R Q - B • S T R U C T U R E + T E X T

GraphSAGE at 70.93% / 0.5183 vs. MLP at 56.12% / 0.3301 → **+14.8 pp** accuracy and **+18.8 points** Macro-F1 with the same text features — the lift is pure graph signal, stable across 2019 and 2020. *Combining structure with text is decisively better than either alone.*

R Q - C • M E C H A N I S M

Spearman ρ = +0.424 (p = 0.0065) → Per-class graph homophily $H(c)$ is positively rank-correlated with per-class GraphSAGE lift $\Delta F1(c)$. The +14.8 pp aggregate boost is concentrated where citation neighborhoods are locally label-coherent. *RQ-A and RQ-B turn out to be two views of one underlying property of the graph.*

The bigger picture — in this domain, graph structure is not a weak auxiliary signal. It is a first-class input. A text-only classifier is leaving a large amount of signal on the table.

Thank you.

Questions?

F u t u r e w o r k d i r e c t i o n s

- Temporal GNNs that respect edge timestamps (e.g., TGN).
- Label-noise-aware training with top-k soft labels for interdisciplinary papers.
- Fairness-aware ranking that controls for publication year.
- Scale the same methodology to OGBN-Papers100M.